

Assignment # 1

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Publisher (PublisherID, PublisherName, Address)

- PublisherID: Integer (Primary Key)
- PublisherName: String
- Address: String

Primary Key: PublisherID

Sample Tuples:-

- (1, "Hassan", "175 Block 67A")
- (2, "Ahmad", "176 Block 67A")
- (3, "Umar", "177 Block 67A")
- (4, "Omar", "178 Block 67A")
- (5, "Farhan", "174 Block A3")

Author (AuthorID, AuthorName, Gender, BirthDate, Country)

- AuthorID: Integer (Primary Key)
- AuthorName: String
- Gender: String
- BirthDate: Date
- Country: String

Primary Key: AuthorID

Sample Tuples:-

- (1, "Hassan", "Male", "1965-07-28", "France")
- (2, "Omar", "Male", "1947-08-16", "Pakistan")
- (3, "Vikram", "Male", "1775-12-16", "America")
- (4, "Firdaus", "Female", "1890-09-15", "Philippines")
- (5, "Taqeer", "Male", "1899-09-16", "Japan")

Book (ISBN, AuthorID, PublisherID,
PublishDate, BookTitle)

- ISBN : String (Primary Key)
- AuthorID : Integer (Foreign Key, references Author, AuthorID)
- PublisherID : Integer (Foreign Key, references Publisher, PublisherID)
- PublishDate : Date
- BookTitle : String

Primary Key : ISBN

Foreign Key : AuthorID (references Author, AuthorID), PublisherID (references Publisher, PublisherID)

Sample Tuples :-

- ("123456", 1, 1, "1997-07-31", "Hamlock")
- ("123455", 2, 2, "2010-11-24", "Harry Potter")
- ("123457", 3, 3, "1813-01-01", "The Ghost")
- ("123458", 4, 4, "1814-01-05", "The Snail")
- ("123459", 5, 1, "1817-08-02", "The Witch")

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Bank Database

Relational Model

Customer (CustomerID, CustomerName, Gender, Contact No, Address, CustomerType)

- CustomerID : Integer (Primary Key)
- CustomerName : string
- Gender : String
- Contact No : string
- Address : String
- CustomerType : String

Primary Key : customerID

Sample Tuples:-

- (1, "Hassan", "Male", "1234567890", "132 Block A7", "Priority")
- (2, "Omar", "Male", "1234567891", "133 Block A7", "Business")
- (3, "Umar", "Male", "1234567892", "123 Block A6", "Corporate")
- (4, "Firdaus", "Female", "1234567893", "123 Block B2", "Corporate")
- (5, "Mohsin", "Male", "1234567894", "124 Block C7", "Priority")

Account (AccountID, CustomerID, AccountType, Balance)

- AccountID : Integer (Primary Key)
- CustomerID : Integer (Foreign Key, references Customer, CustomerID)
- Account Type : String
- Balance : Decimal

Primary Key: AccountID

Foreign Key: CustomerID References Customer.CustomerID

Sample Types:-

- (1, 1, "Saving", 1000.00)
- (2, 2, "Current", 5000.00)
- (3, 3, "current", 10000.00)
- (4, 4, "Saving", 8000.00)
- (5, 5, "current", 2000.00)

Transaction (TransactionID, AccountID, TransactionDate, Amount)

- TransactionID: Integer (Primary Key)
- AccountID: Integer (Foreign Key, references Account.AccountID)
- TransactionDate: Date
- Amount: Decimal

Primary Key: TransactionID

Foreign Key: AccountID References Account.AccountID

Sample Types:-

- (1, 1, "2022-01-02", 100.00)
- (2, 2, "2022-02-01", 500.00)
- (3, 3, "2023-03-02", 1000.00)
- (4, 4, "2022-04-05", 200.00)
- (5, 5, "2023-05-07", 150.00)

Product (ProductID, ProductName, ProductDescription, ProductCategory)

- ProductID: Integer (Primary Key)
- ProductName: String
- ProductDescription: String
- ProductCategory: String

Primary Key: ProductID

Sample Tuples:-

- (1, "Personal Loan", "Loan for personal use", "Personal Loan")
- (2, "Car Loan", "Loan for purchasing a car", "Car Loan")
- (3, "Mortgage Loan", "Loan for purchasing a property", "Mortgage Loan")
- (4, "Education Loan", "Loan for education", "Education Loan")
- (5, "Business Loan", "Loan for business", "Business Loan")

Constraints:-

Each Customer can have only one account (enforced by the primary key constraint on Customer)

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SuperKeys:- $\{A, B, C, D\}, \{A, B, C\}, \{A, B, D\}, \{A, C, D\}, \{A, B, D\}, \{B, C, D\}, \{A, C\}, \{B, C\}, \{A, D\}, \{B, D\}, \{C, D\}, \{C\}$

Now we will choose all the candidate keys (Super keys whose proper subset is not a super key or it is a MINIMAL super key)

Candidate Keys:- ~~{A, B, C}~~, {A, D},
{B, D}, {C}

There are ~~4~~³ candidate keys for this current state of relation.

4 1

a. The INSERT INTO Product VALUES (null, 'Chicken Nuggets', 120, 100); operation will not be successful, it will fail because the ProductID column in Product table is defined as a primary key and cannot contain null values. Therefore, a null value cannot be inserted into this column. This operation violates the NOT NULL constraint of the ProductID column.

b. The INSERT INTO Production VALUES (1, '1-2001', 300, '01-Feb-2021', '01-Feb-2022'); operation will be successful. It will add a new row in the Production table with ProductID = 1, BatchNo = '1-2001', UnitsProduced = 300, DateOfProduction = '01-Feb-2021', and ExpiryDate = '01-Feb-2022'. This operation does not violate any integrity constraints.

c. INSERT INTO Production VALUES (1, null, 300, '01-Feb-2021', '01-Feb-2022'); This operation would violate the integrity constraint as the BatchNo column in the Production table is a NOT NULL column. Therefore, this operation cannot be carried out.

d) `INSERT INTO customer VALUES (1, 'Packaged', 'DHA', '042-31234567');` This operation would be carried out successfully. The database after this operation would be:

Customer

CustomerID CustomerName Address ContactNo

1	Metro	Model Town	042-31234567
2	Jalal Sons	Gulberg	042-31234568
3	Rainbow	Bahria Town	042-31234569
4	Packages	DHA	042-31234567

e) `INSERT INTO Order VALUES (5, 10, 1, '09-Feb-2021');` This operation would violate the integrity constraint as the CustomerID 10 does not exist in the Customer table. Therefore, this operation cannot be carried out.

2
a) Successful operation. The row with ProductID = 2 will be deleted from the Product table. No integrity constraints will be violated.

b) Successful operation. The row with CustomerID = 1 will be deleted from the Customer table and all rows from the Order table where CustomerID = 1 will be automatically deleted due to the CASCADE option. No integrity constraints will be violated.

c. Successful operation. The row(s) with BatchID = '1-1002' will be deleted from the Production table. No integrity constraints will be violated.

d. Successful operation: All rows from the Order table where CustomerID = 1 or ProductID = 1 will be deleted. No integrity constraints will be violated.

e. Unsuccessful operation, There is no row in the Order table with orderNo = 4, so nothing will be deleted.

3

a This operation violates the foreign key constraint in the Order table because ProductID = 2 is referenced by order table as a foreign key. However, since SET NULL option is implemented, the referenced foreign key column in Order table will be updated to NULL. But this is not possible since CustomerID cannot be NULL in order table. Therefore, this operation will fail.

b. This operation violates the foreign key constraint in the order table because CustomerID = 1 is referenced by order table as a foreign key. However, since SET NULL option is implemented, the referenced foreign key column in order table will be updated to NULL. But this is not possible since ProductID cannot be NULL in order table, therefore this operation will fail.

c. This operation will succeed as there is no other table referencing Production.BatchID as a foreign key.

d. This operation violates the foreign key constraint in the Production table because ProductID = 1 is referenced by Production table as a foreign key. However, since SETNULL option is implemented, the referenced foreign key column in Production table will be updated to NULL. But this is not possible since ProductID cannot be NULL in Production table. Therefore, this operation will fail.

e. This operation will fail as there is no Order with OrderNo = 4.

III

a. Successful operation: This will delete the row from the Product table where ProductID = 2.

b. Unsuccessful operation: This operation cannot be carried out because there are related rows in the Order table with CustomerID = 1. The foreign key constraint between Order, CustomerID and Customer, CustomerID will prevent the deletion of the row from the Customer table.

c. Successful operation: This will delete the row from the Production table where BatchID = '1-1002'.

d. Unsuccessful operation: This operation cannot be carried out because there are rows related in the Production table with ProductID = 1 and the foreign key constraint between Production, ProductID and Product, ProductID is set to NO ACTION. The constraint prevents the deletion of the rows from the Order table.

e

The operation will be successful. The operation will not remove any rows from the table, but it will also not encounter any errors or integrity constraint violations. The database will remain unchanged after this operation.

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a. This operation will be carried out successfully as it ~~will violate the~~ foreign key with CASCADE option is implemented. The changes made to the database will be that the Customer ID of order with Order No = 2 will be updated from 2 to 10.

b. This operation will not be carried out successfully as a foreign key with CASCADE option is implemented. The changes made to the database will be that the OrderDate of orders with ProductID = 1 will be set to NULL.

c. This operation will be carried out successfully as a foreign key with CASCADE option is implemented. The changes made to the database will be that the OrderDate of orders with ProductID = 1 will be set to NULL.

d. This operation will not be carried out successfully as it will violate the foreign key constraint in the Product table. The changes will not be made to database.

e. This operation will be carried out successfully as a foreign key with CASCADE option is implemented. The changes made to the database will be that the BatchNo of the Production record with UnitsProduced = 6000 will be set to NULL.

6.

a. This operation would be carried out successfully. The row in the Order table where CustomerID is 2 will be updated to set CustomerID to 10. Since the foreign key with SET NULL option is implemented, the rows in other tables that reference the updated row will have their foreign key values set to NULL, which in this case is none.

b. This operation would violate the integrity constraint of referential integrity, since the ProductID in the Production and Order tables reference the ProductID in the Product table. If we were to update the ProductID in the Product table to 21, then the rows in the Production and Order tables that reference the ProductID of 3 would no longer reference a valid row in the Product table. Therefore, this operation would not be carried out successfully.

c. This operation would be carried out successfully. The row in the Order table where ProductID is 1 will be updated to set OrderDate to NULL. Since the foreign key with SET NULL option is implemented, the rows in other tables that reference the updated row will have their foreign key values set to NULL, which in this case is none.

d. This operation would violate the integrity constraint of referential integrity, since the CustomerID in the Production table references the CustomerID in the Customer table. If we were to update the CustomerID in the Production table to 1, then the rows in the Production table that reference the original CustomerID would no longer reference a valid row in the Customer table. Therefore, this operation would not be carried out successfully.

e. This operation would be carried out successfully. The row in the Production table where UnitsProduced is 6000 will be updated to set BatchNo to NULL. Since the foreign key with SET NULL option is implemented, the rows in other tables that reference the updated row will have their foreign key values set to NULL, which in this case is none.

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a. This operation would be carried out successfully because the foreign key constraint on CustomerID in the Order table has the NO ACTION option, which means that the update operation will not be prevented even if it results in a violation of the constraint. However, the integrity constraint is violated, as the updated CustomerID value does not exist in the Customer table.

b. This operation would be carried out successfully. Since the ProductID is not used as a foreign key in any other table, updating the ProductID would not violate any foreign key constraints.

c. This operation would be carried out successfully. Since OrderDate is a nullable attribute, setting it to NULL would not violate any constraints.

d. This operation would be prevented by the foreign key constraint on CustomerID in the Production table, which has the NO ACTION option. The CustomerID value being updated to 1 does not exist in the Customer table. Thus, this operation would result in a violation of the foreign key constraint.

e. This operation would be carried out successfully. Since BatchNo is a nullable attribute, setting it to NULL would not violate any constraints.

5

SQL provides several date and time formats and functions for working with date and time values.

Date and Time Formats:-

1. YYYY-MM-DD: A standard date format, where YYYY represents the 4-digit year, MM represents the 2-digit month, and DD represents the 2-digit ~~hour~~ day.
2. HH:MM:SS: A standard time format, where HH represents the 2-digit hour, MM represents the 2-digit minute, and SS represents the 2-digit second.
3. YYYY-MM-DD HH:MM:SS: A standard date and time format, combining the above two formats.

4. DD/MM/YYYY: A different format for specifying dates, where DD represents the 2-digit day, MM represents the 2-digit month, and YYYY represents the 4-digit year.

Date and Time Functions:

1. CURRENT_DATE: Returns the current date
2. CURRENT_TIME: Returns the current time
3. CURRENT_TIMESTAMP: Returns the current date and time.
4. DATE: Extracts the date from a date-time expression.
5. TIME: Extracts the time from a date-time expression.
6. YEAR: Extracts the year from a date-time expression.
7. MONTH: Extracts the month from a date-time expression.
8. DAY: Extracts the day of the month from a date-time expression.
9. HOUR: Extracts the hour from a date-time expression.
10. MINUTE: Extracts the minute from a date-time expression.
11. SECOND: Extracts the second from a date-time expression.
12. NOW: Returns the current date and time.
13. DATEDIFF: Calculates the difference between two dates.
14. ADDDATE: Adds a specified number of ~~dates~~ days to a date.
15. SUBDATE: Subtracts a specified number of days from a date.
16. DATE_ADD: Adds a specified time interval to a date-time expression.
17. DATE_SUB: Subtracts a specified time interval from a date-time expression.

16

Composite primary key and composite foreign key are two concepts in database design that involve using multiple attributes to create a unique identifier for entities or relationships in a database.

A composite primary key is a primary key that is made up of two or more attributes. In other words, a composite primary key is a key that is formed by combining two or more columns to create a unique identifier for each tuple in a relation. For example, consider a relation "Order" with the attributes "OrderID", "CustomerID", and "OrderDate". If the primary key for this relation is "OrderID", then this would be a simple primary key. However, if the primary key for this relation is "OrderID" and "CustomerID" combined, then this would be a composite primary key, since the combination of these two attributes will uniquely identify each tuple in the relation.

A composite foreign key is a foreign key that is made up of two or more attributes. In other words, a composite foreign key is a key that is formed by combining two or more columns to create a relationship between two relations. For example, consider two relations "Order" and "Customer", where "Order" has a composite primary key of "CustomerID". In this scenario, "CustomerID" would be a composite foreign key in the "Order" relation, since it references the primary key of the "Customer" relation, and "OrderID" would be part of the composite foreign key as well.

Q / Part No.

In Summary, composite primary key and composite foreign key are used to create unique identifiers for entities or relationships in a database, and to establish relationships between entities in a database, respectively.
